

DSA0603 – Data Handling and Visualization

CO2 – Assessment Test 5 (AT2)

Visualization Development Exercise — Questions with Answers

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Question 1 — Chart Selection (Topic 1: Directory of Visualizations)

For each scenario, the single best chart type and justification are given below.

1. *Comparing total revenue across 6 product categories in one year.*

Best chart: Bar Chart (Amounts)

Justification: A bar chart uses length to encode magnitude, which the human eye compares very accurately. With only 6 discrete categories, a bar chart makes it easy to rank categories and compare exact revenue values side by side.

2. *Showing how the salary distribution differs across 4 job levels.*

Best chart: Box Plot / Violin Plot (Distribution)

Justification: Box or violin plots summarize the median, quartiles, spread, and outliers of a continuous variable for each group. Placing 4 box/violin plots side by side lets us compare the shape and spread of salaries across job levels directly.

3. *Showing market share of 5 competing brands.*

Best chart: Pie Chart (Proportion)

Justification: Market share values are parts of a whole (they sum to 100%). With only 5 categories, a pie chart clearly communicates each brand's share of the total. (A 100% stacked bar is a reasonable alternative if more precise value comparison is needed.)

4. *Showing the relationship between advertising spend and sales revenue.*

Best chart: Scatter Plot (x-y Relationship)

Justification: A scatter plot places one continuous variable on each axis, making it the standard choice for revealing correlation, trend, and outliers between two numeric variables such as spend and revenue.

5. *Showing monthly website traffic over 3 years.*

Best chart: Line Chart (Temporal)

Justification: Line charts connect ordered points over time, making trends, seasonality, and growth/decline patterns across 36 months immediately visible in a single continuous view.

6. *Showing which employees report to which managers in an org.*

Best chart: Node-Link / Tree (Hierarchy) Diagram (Network)

Justification: Reporting relationships are a hierarchical network of connections between people. A node-link (org-chart/tree) diagram explicitly draws each employee-manager edge, making the reporting structure easy to trace.

Question 2 — Bar Plot of Average Score by Subject

Dataset: avg_score.csv (Subject, Average_Score). Task: vertical bar plot sorted descending, single consistent bar color with the top scorer highlighted, and data labels on each bar.

Python Implementation

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "Subject": ["Mathematics", "Science", "English", "History", "Computer Sci"],
    "Average_Score": [72, 68, 75, 64, 81]
})

# Sort descending by score
df = df.sort_values("Average_Score", ascending=False).reset_index(drop=True)

# Single consistent color; highlight the top scorer
colors = ["#4C72B0"] * len(df)
max_idx = df["Average_Score"].idxmax()
colors[max_idx] = "#DD8452"

fig, ax = plt.subplots(figsize=(8, 5.5))
bars = ax.bar(df["Subject"], df["Average_Score"], color=colors,
              edgecolor="black", linewidth=0.6)

# Data labels on top of each bar
for bar, val in zip(bars, df["Average_Score"]):
    ax.text(bar.get_x() + bar.get_width()/2, val + 1, str(val),
            ha="center", va="bottom", fontsize=11, fontweight="bold")

ax.set_title("Average Score by Subject (Descending Order)")
ax.set_xlabel("Subject")
ax.set_ylabel("Average Score")
plt.tight_layout()
plt.savefig("q2_bar.png", dpi=150)
plt.show()
```

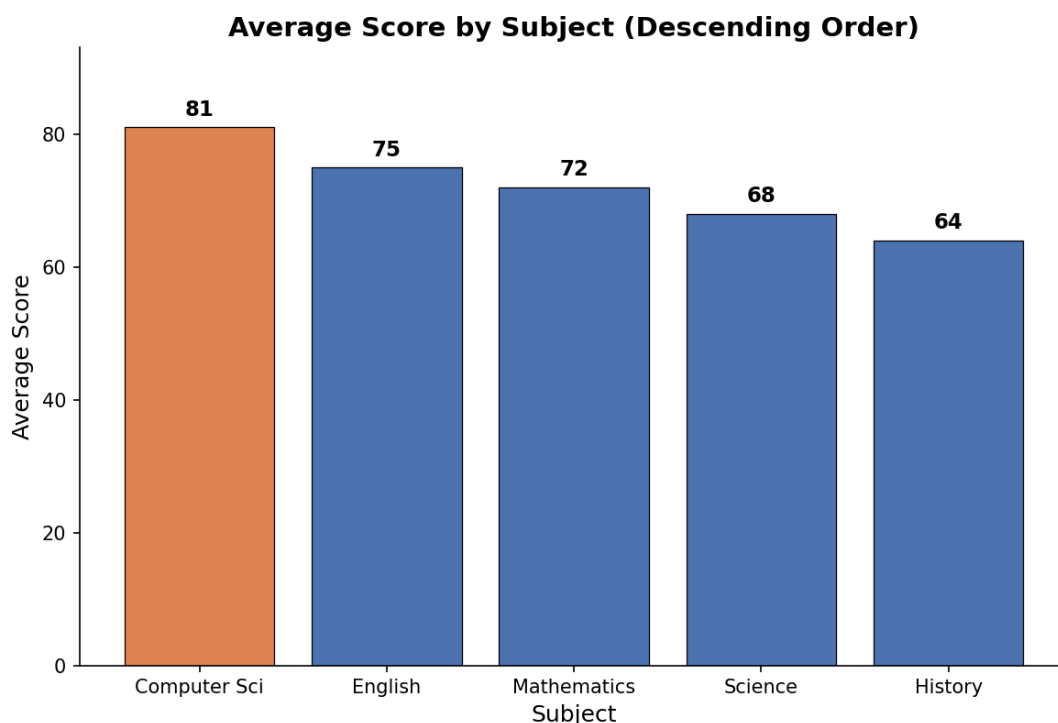


Figure Q2 — Average Score by Subject (Python, matplotlib)

R Implementation

```
library(ggplot2)
library(dplyr)

df <- data.frame(
  Subject = c("Mathematics", "Science", "English", "History", "Computer Sci"),
  Average_Score = c(72, 68, 75, 64, 81)
)

# Sort descending and mark the highest scorer
df <- df %>%
  arrange(desc(Average_Score)) %>%
  mutate(Subject = factor(Subject, levels = Subject),
         Highlight = ifelse(Average_Score == max(Average_Score), 'Top', 'Other'))

ggplot(df, aes(x = Subject, y = Average_Score, fill = Highlight)) +
  geom_col(color = 'black', width = 0.7) +
  geom_text(aes(label = Average_Score), vjust = -0.5, fontface = 'bold') +
  scale_fill_manual(values = c('Top' = '#DD8452', 'Other' = '#4C72B0')) +
  labs(title = 'Average Score by Subject (Descending Order)',
       x = 'Subject', y = 'Average Score') +
  theme_minimal() +
  theme(legend.position = 'none')

ggsave("q2_bar_R.png", width = 8, height = 5.5, dpi = 150)
```

Interpretation: Computer Science has the highest average score (81), highlighted in orange, while History has the lowest (64). Because the bars are sorted in descending order, the ranking of subjects by performance is immediately visible without needing to read every label; Mathematics, Science and English cluster in the middle 68–75 range.

Question 3 — Regional Sales: Four Views of the Same Data

Dataset: regional_sales.csv (Region, Quarter, Sales_000s). Four different chart types are produced from the same underlying data to compare how each view highlights different aspects of the data.

Python Implementation (all four charts)

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

data = {
    "Region": ["North"]*4 + ["South"]*4 + ["East"]*4 + ["West"]*4,
    "Quarter": ["Q1", "Q2", "Q3", "Q4"]*4,
    "Sales_000s": [120, 135, 128, 150, 95, 110, 105, 130,
                  140, 138, 145, 160, 80, 85, 90, 100]
}
df = pd.DataFrame(data)

# Reshape: Region x Quarter pivot table
pivot = df.pivot_table(index="Region", columns="Quarter", values="Sales_000s")
pivot = pivot.loc[["North", "South", "East", "West"]]

# 1. Grouped bar chart
x = np.arange(len(pivot.index)); width = 0.2
quarters = ["Q1", "Q2", "Q3", "Q4"]
fig, ax = plt.subplots(figsize=(9, 5.5))
for i, q in enumerate(quarters):
    ax.bar(x + (i - 1.5)*width, pivot[q], width, label=q)
ax.set_xticks(x); ax.set_xticklabels(pivot.index)
ax.set_title("Grouped Bar: Sales by Region and Quarter")
ax.legend(title='Quarter'); plt.tight_layout()
plt.savefig("q3_grouped_bar.png", dpi=150)

# 2. Stacked bar chart
fig, ax = plt.subplots(figsize=(9, 5.5))
bottom = np.zeros(len(pivot))
for q in quarters:
    ax.bar(pivot.index, pivot[q], bottom=bottom, label=q)
    bottom += pivot[q].values
ax.set_title("Stacked Bar: Sales by Region, Stacked by Quarter")
ax.legend(title='Quarter'); plt.tight_layout()
plt.savefig("q3_stacked_bar.png", dpi=150)

# 3. Dot plot: total annual sales per region, ranked
totals = df.groupby("Region")["Sales_000s"].sum().sort_values(ascending=False)
fig, ax = plt.subplots(figsize=(8, 5))
y_pos = np.arange(len(totals))
ax.hlines(y=y_pos, xmin=0, xmax=totals.values, color="grey", alpha=0.5)
ax.plot(totals.values, y_pos, "o", markersize=12, color="#C44E52")
ax.set_yticks(y_pos); ax.set_yticklabels(totals.index); ax.invert_yaxis()
ax.set_title("Dot Plot: Total Annual Sales by Region (Ranked)")
plt.tight_layout()
plt.savefig("q3_dot_plot.png", dpi=150)

# 4. Heatmap
fig, ax = plt.subplots(figsize=(7.5, 5.5))
sns.heatmap(pivot[quarters], annot=True, fmt=".0f", cmap="YlOrRd",
            cbar_kws={"label": "Sales (000s)"})
ax.set_title("Heatmap: Sales by Region and Quarter")
```

```
plt.tight_layout()
plt.savefig("q3_heatmap.png", dpi=150)
```

1. Grouped Bar Chart

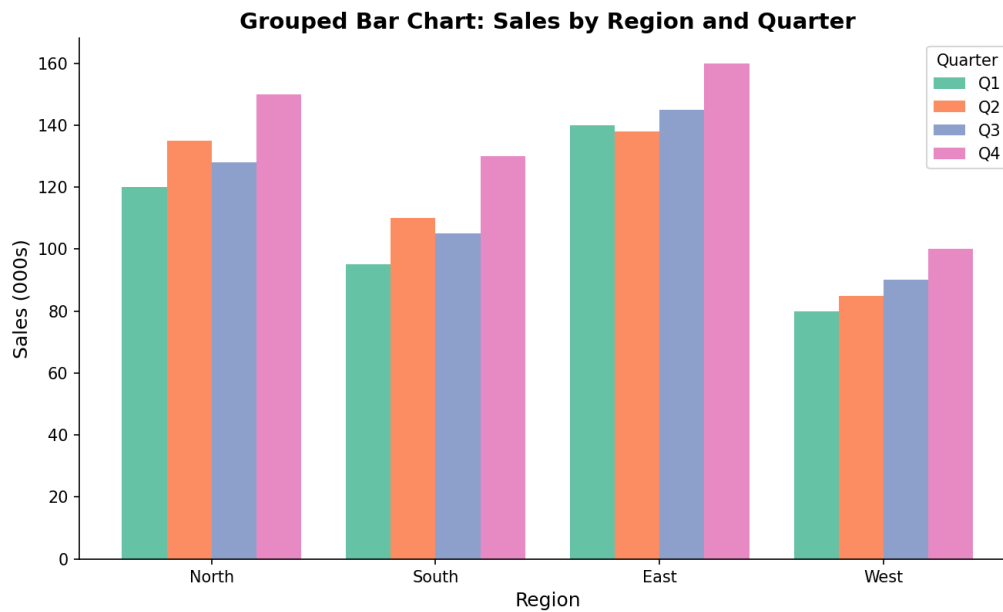


Figure Q3.1 — Grouped Bar Chart: Sales by Region and Quarter

Interpretation: East leads in almost every quarter and West lags behind throughout the year. Most regions show a rising trend from Q1 to Q4, with North dipping slightly in Q3 before rebounding strongly in Q4. The grouped layout makes quarter-to-quarter comparison within each region easy, but comparing total regional performance requires mental addition.

2. Stacked Bar Chart

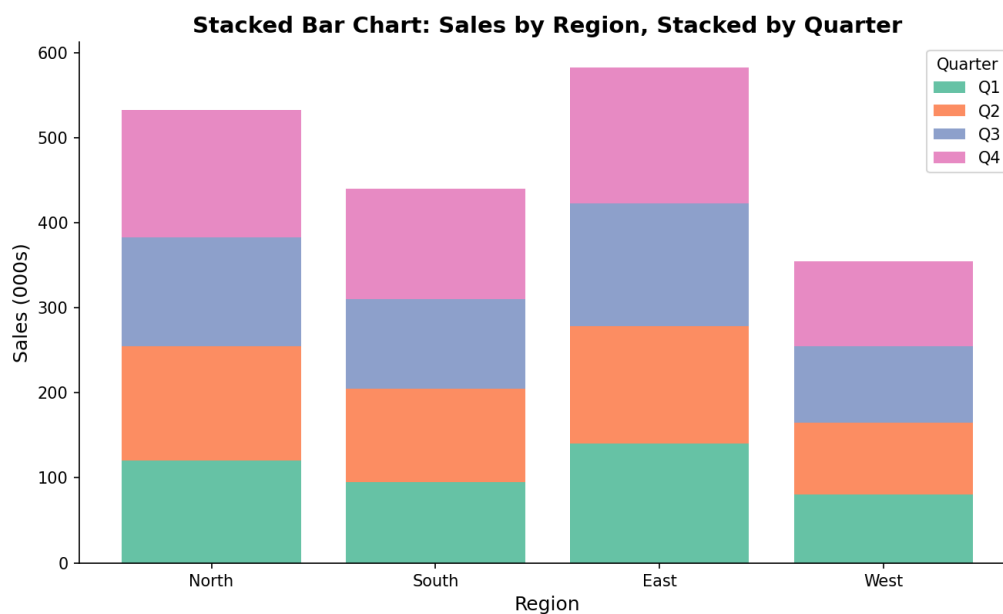


Figure Q3.2 — Stacked Bar Chart: Sales by Region, Stacked by Quarter

Interpretation: The stacked view emphasizes total annual sales per region through overall bar height: East's bar is tallest and West's is shortest. However, comparing the size of a single quarter's segment (e.g., Q2 across regions) is harder here than in the grouped chart, since segments do not share a common baseline except the bottom one.

3. Dot Plot (Total Annual Sales, Ranked)

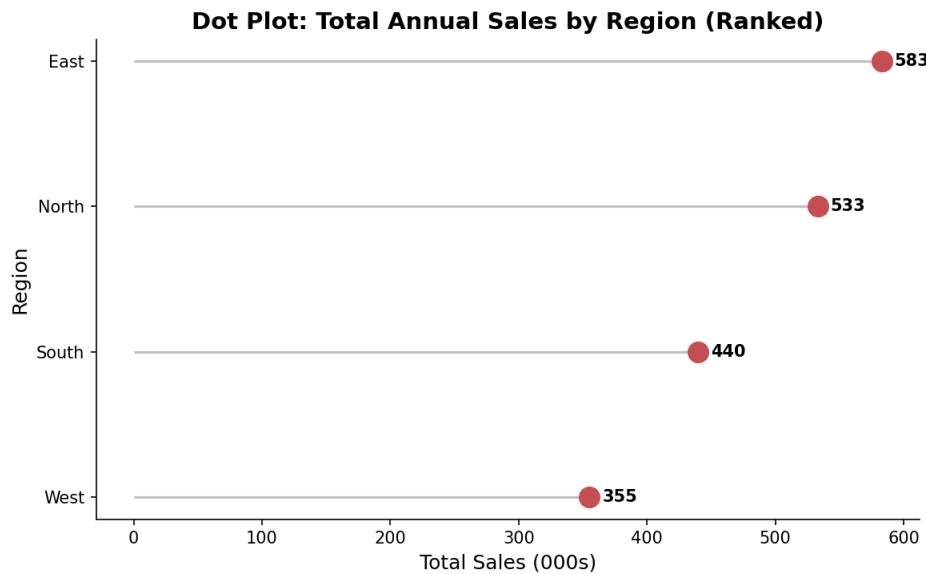


Figure Q3.3 — Dot Plot: Total Annual Sales by Region (Ranked)

Interpretation: Ranked from highest to lowest: East (583k) > North (533k) > South (440k) > West (355k). The dot plot removes the visual clutter of bar area and color segments, making the ranking and the magnitude gaps between regions (e.g., the ~123k gap between South and West) very easy to read at a glance.

4. Heatmap

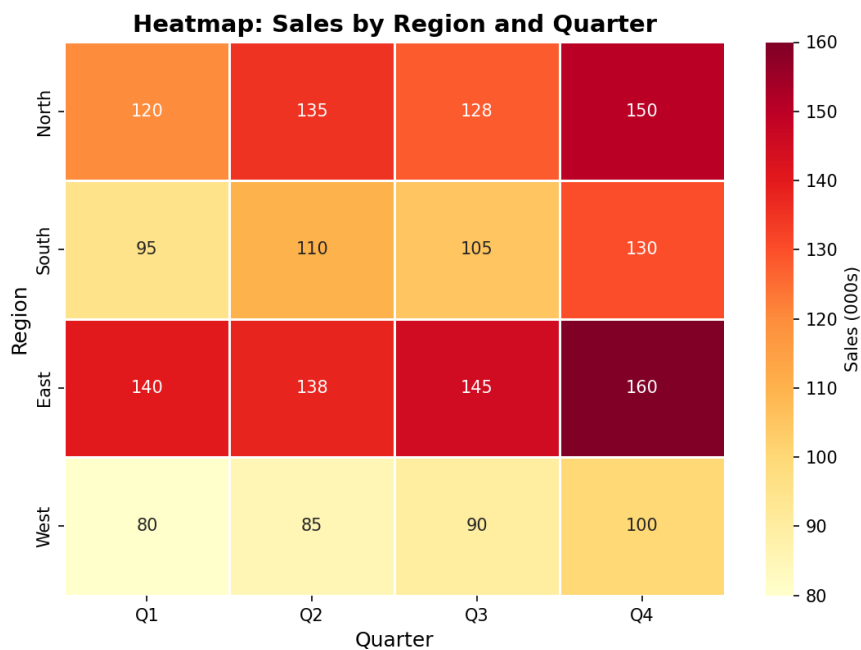


Figure Q3.4 — Heatmap: Region x Quarter, Color-Encoded by Sales

Interpretation: The heatmap shows Q4 as the darkest column for every region, confirming a company-wide seasonal peak at year-end. West is the lightest row throughout, confirming it is consistently the weakest-performing region in every quarter, not just in total. The heatmap is best for spotting cell-level patterns across two categorical dimensions simultaneously, something the other three charts cannot do as compactly.

R Implementation (all four charts)

```
library(tidyr)
library(dplyr)
library(ggplot2)

df <- data.frame(
  Region = rep(c('North', 'South', 'East', 'West'), each = 4),
  Quarter = rep(c('Q1', 'Q2', 'Q3', 'Q4'), times = 4),
  Sales_000s = c(120, 135, 128, 150, 95, 110, 105, 130,
                 140, 138, 145, 160, 80, 85, 90, 100)
)
df$Region <- factor(df$Region, levels = c('North', 'South', 'East', 'West'))

# 1. Grouped bar chart
ggplot(df, aes(x = Region, y = Sales_000s, fill = Quarter)) +
  geom_col(position = 'dodge') +
  labs(title = 'Grouped Bar: Sales by Region and Quarter') +
  theme_minimal()
ggsave('q3_grouped_bar_R.png', width = 9, height = 5.5)

# 2. Stacked bar chart
ggplot(df, aes(x = Region, y = Sales_000s, fill = Quarter)) +
  geom_col(position = 'stack') +
  labs(title = 'Stacked Bar: Sales by Region, Stacked by Quarter') +
  theme_minimal()
ggsave('q3_stacked_bar_R.png', width = 9, height = 5.5)

# 3. Dot plot: total annual sales per region, ranked
totals <- df %>%
  group_by(Region) %>%
  summarise(Total = sum(Sales_000s)) %>%
  arrange(desc(Total)) %>%
  mutate(Region = factor(Region, levels = rev(Region)))

ggplot(totals, aes(x = Total, y = Region)) +
  geom_segment(aes(x = 0, xend = Total, y = Region, yend = Region), color = 'grey60') +
  geom_point(size = 5, color = '#C44E52') +
  labs(title = 'Dot Plot: Total Annual Sales by Region (Ranked)') +
  theme_minimal()
ggsave('q3_dot_plot_R.png', width = 8, height = 5)

# 4. Heatmap
ggplot(df, aes(x = Quarter, y = Region, fill = Sales_000s)) +
  geom_tile(color = 'white') +
  geom_text(aes(label = Sales_000s)) +
  scale_fill_gradient(low = 'lightyellow', high = 'darkred') +
  labs(title = 'Heatmap: Sales by Region and Quarter') +
  theme_minimal()
ggsave('q3_heatmap_R.png', width = 7.5, height = 5.5)
```

Question 4 — Exam Score Distributions by Class

Synthetic data: Class A $\sim N(70, 8)$, Class B $\sim N(75, 5)$, Class C $\sim N(65, 12)$, Class D $\sim N(80, 6)$, 40 scores each, seed = 42 (used identically in Python's `numpy.random.seed` and R's `set.seed` for consistent results).

Python Implementation

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

np.random.seed(42)
classA = np.random.normal(70, 8, 40)
classB = np.random.normal(75, 5, 40)
classC = np.random.normal(65, 12, 40)
classD = np.random.normal(80, 6, 40)

df = pd.DataFrame({
    "Score": np.concatenate([classA, classB, classC, classD]),
    "Class": ["A"]*40 + ["B"]*40 + ["C"]*40 + ["D"]*40
})

# 1. Violin plot with embedded box/median
fig, ax = plt.subplots(figsize=(8, 5.5))
sns.violinplot(data=df, x="Class", y="Score", order=["A", "B", "C", "D"],
               inner="box", palette="Set2")
ax.set_title("Exam Score Distribution by Class (Violin Plot)")
plt.tight_layout()
plt.savefig("q4_violin.png", dpi=150)

# 2. Ridgeline plot, ordered by mean score
means = df.groupby("Class")["Score"].mean().sort_values(ascending=False)
order = means.index.tolist()
fig, axes = plt.subplots(len(order), 1, figsize=(8, 6), sharex=True)
for i, cls in enumerate(order):
    vals = df.loc[df["Class"] == cls, "Score"]
    sns.kdeplot(vals, ax=axes[i], fill=True)
    axes[i].set_ylabel("")
plt.subplots_adjust(hspace=-0.1)
plt.savefig("q4_ridgeline.png", dpi=150)
```

1. Violin Plot (with embedded boxplot)

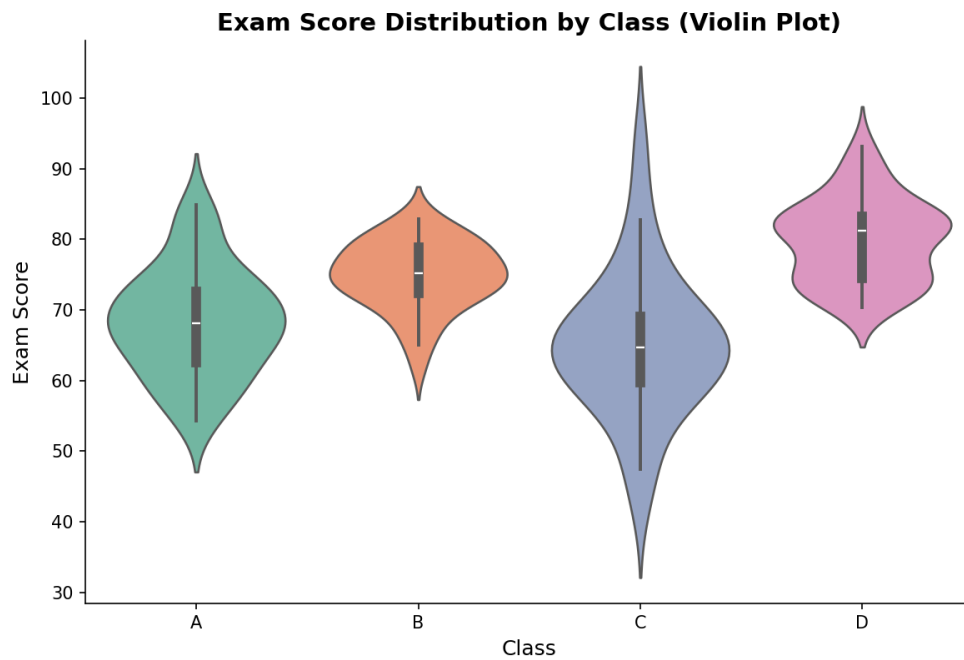


Figure Q4.1 — Exam Score Distribution by Class (Violin Plot)

Interpretation: Class D shows the highest median (~ 80) and a relatively narrow, tall violin, reflecting its lower standard deviation ($sd = 6$) — most students in Class D score consistently well. Class C has the widest violin, consistent with its high standard deviation ($sd = 12$), indicating a much more variable range of performance from weak to strong students. Class B is tightly clustered around its mean (~ 75), the most consistent class overall.

2. Ridgeline Plot (ordered by mean score)

Ridgeline Plot: Exam Score Density by Class (Ordered by Mean)

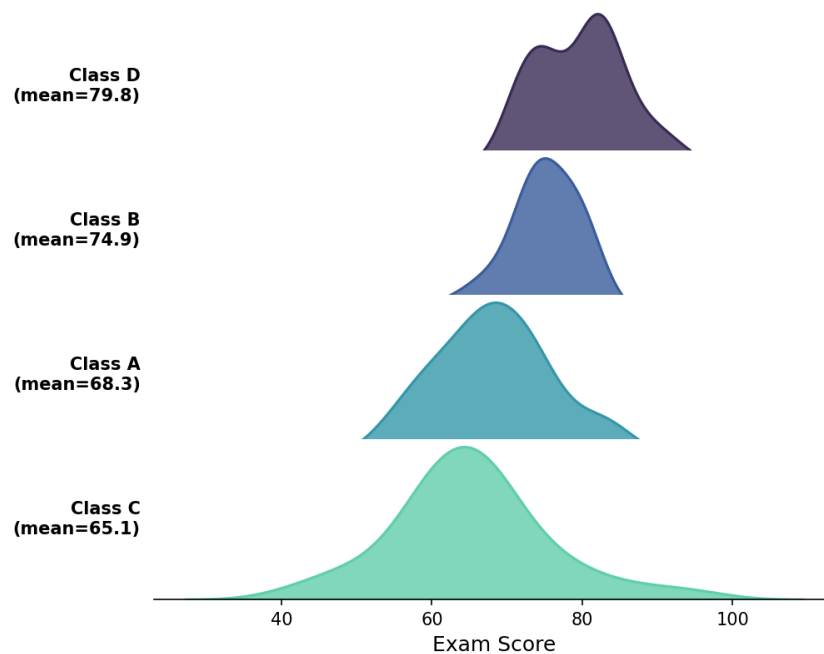


Figure Q4.2 — Ridgeline Plot: Exam Score Density by Class (Ordered by Mean)

Interpretation: Ordering the ridges by mean score ($D=79.8 > B=74.9 > A=68.3 > C=65.1$) makes the class ranking visually obvious from top to bottom. Class C's curve is clearly the flattest and widest, reinforcing the high variability seen in the violin plot, while Class D's curve is the narrowest and shifted furthest right, confirming both a high mean and low spread.

R Implementation

```
library(ggplot2)
library(ggribes)
library(dplyr)

set.seed(42)
classA <- rnorm(40, mean = 70, sd = 8)
classB <- rnorm(40, mean = 75, sd = 5)
classC <- rnorm(40, mean = 65, sd = 12)
classD <- rnorm(40, mean = 80, sd = 6)

df <- data.frame(
  Score = c(classA, classB, classC, classD),
  Class = rep(c('A','B','C','D'), each = 40)
)

# 1. Violin plot with embedded boxplot
ggplot(df, aes(x = Class, y = Score, fill = Class)) +
  geom_violin(trim = FALSE) +
  geom_boxplot(width = 0.1, fill = 'white') +
  labs(title = 'Exam Score Distribution by Class (Violin Plot)') +
  theme_minimal() + theme(legend.position = 'none')
ggsave('q4_violin_R.png', width = 8, height = 5.5)

# 2. Ridgeline plot, ordered by mean score
class_order <- df %>%
  group_by(Class) %>%
  summarise(m = mean(Score)) %>%
  arrange(m) %>%
  pull(Class)
df$Class <- factor(df$Class, levels = class_order)

ggplot(df, aes(x = Score, y = Class, fill = Class)) +
  geom_density_ridges(alpha = 0.8) +
  labs(title = 'Ridgeline Plot: Exam Score Density by Class') +
  theme_minimal() + theme(legend.position = 'none')
ggsave('q4_ridgeline_R.png', width = 8, height = 6)
```

Question 5 — Histogram and KDE of Total Bill (tips dataset)

Dataset: seaborn's built-in tips dataset (R equivalent via the reshape2 package). Continuous variable analyzed: total_bill.

Python Implementation

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")

# 1 & 2. Histogram (30 bins) with KDE overlay - overall
fig, ax = plt.subplots(figsize=(8, 5.5))
sns.histplot(tips["total_bill"], bins=30, kde=True, color="#4C72B0")
ax.set_title("Distribution of Total Bill (Histogram + KDE)")
plt.tight_layout()
plt.savefig("q5_hist_overall.png", dpi=150)

# 3. Grouped by smoker
fig, ax = plt.subplots(figsize=(8, 5.5))
sns.histplot(data=tips, x="total_bill", hue="smoker", bins=30, kde=True,
             stat="density", common_norm=False, alpha=0.5)
ax.set_title("Distribution of Total Bill by Smoker Status")
plt.tight_layout()
plt.savefig("q5_hist_by_smoker.png", dpi=150)
```

1 & 2. Histogram (30 bins) with KDE Overlay — Overall

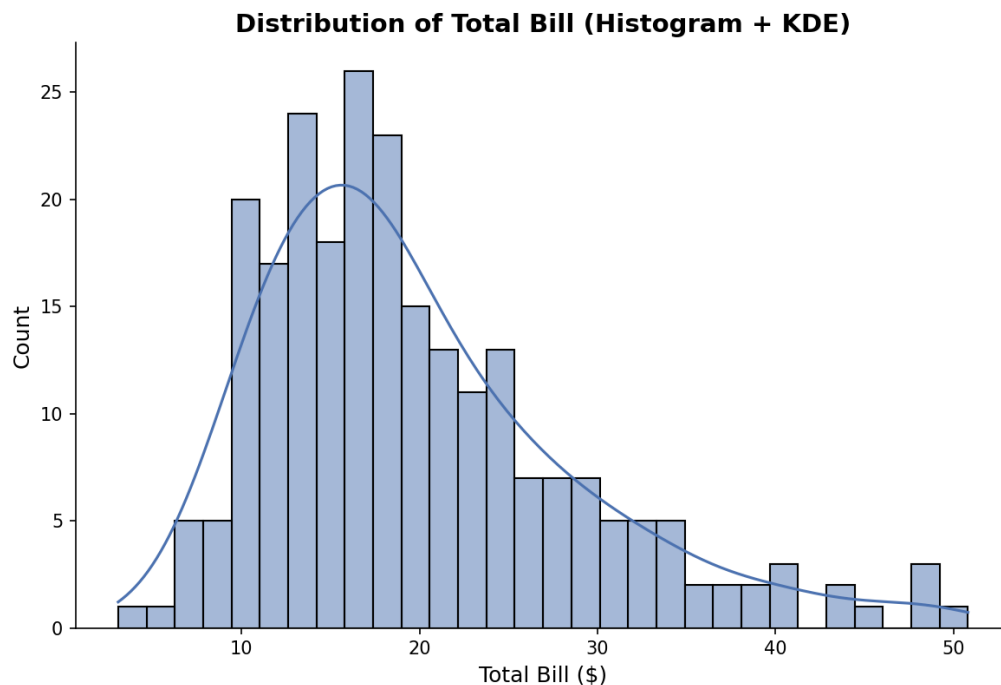


Figure Q5.1 — Distribution of Total Bill (Histogram + KDE)

Interpretation: The distribution of total_bill is unimodal and right-skewed: most bills fall between roughly \$10 and \$20, with a peak (mode) around \$15-18, and a long tail of larger bills extending past \$40. The KDE curve smooths the bar-to-bar noise of the histogram and makes this skew and the single peak easier to see than the raw bar heights alone.

3. Histogram + KDE Grouped by Smoker

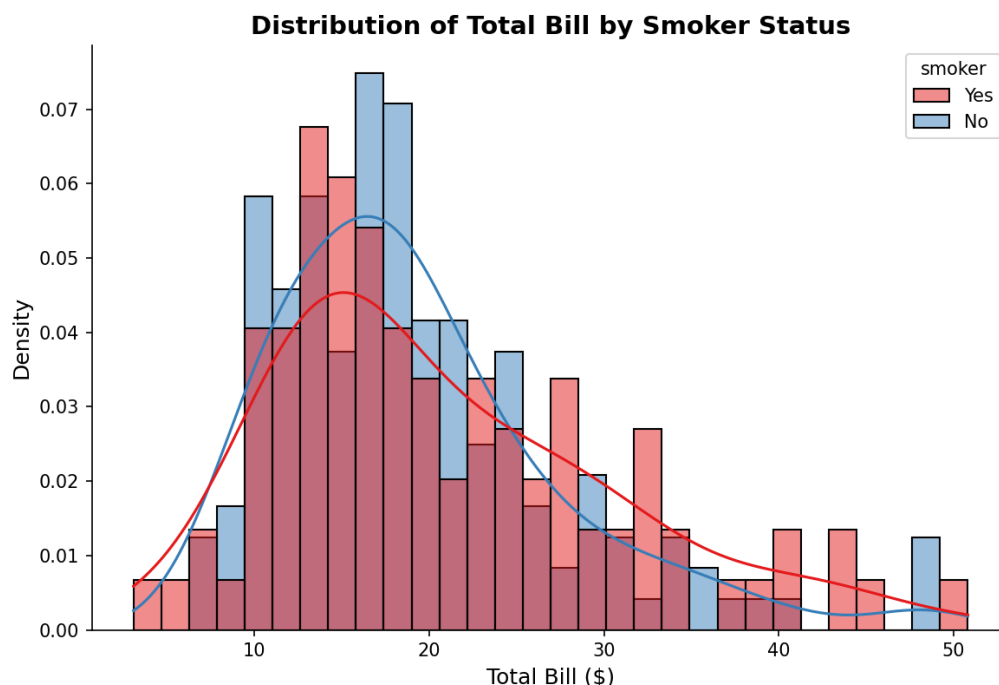


Figure Q5.2 — Distribution of Total Bill by Smoker Status

Interpretation: Smokers have a slightly higher mean total bill (\$20.76) than non-smokers (\$19.19), and the smoker density shows a heavier, more spread-out tail toward larger bills, while the non-smoker density is more concentrated with a sharper peak around \$15-18. Both groups remain right-skewed, but the smoker group's greater spread (std ~9.83 vs ~8.26) suggests more variable spending behavior among smokers.

R Implementation

```
library(ggplot2)
library(reshape2) # provides the `tips` dataset

data(tips)

# 1 & 2. Histogram (30 bins) with KDE overlay - overall
ggplot(tips, aes(x = total_bill)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30,
                 fill = '#4C72B0', color = 'black', alpha = 0.7) +
  geom_density(color = 'darkred', linewidth = 1) +
  labs(title = 'Distribution of Total Bill (Histogram + KDE)',
       x = 'Total Bill ($)', y = 'Density') +
  theme_minimal()
ggsave('q5_hist_overall_R.png', width = 8, height = 5.5)

# 3. Grouped by smoker
ggplot(tips, aes(x = total_bill, fill = smoker)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30,
                 position = 'identity', alpha = 0.5) +
  geom_density(aes(color = smoker), linewidth = 1) +
  labs(title = 'Distribution of Total Bill by Smoker Status',
       x = 'Total Bill ($)', y = 'Density') +
  theme_minimal()
ggsave('q5_hist_by_smoker_R.png', width = 8, height = 5.5)
```